FreelanceSafe Lite

Milestone Escrow, Rule-Based Risk Scoring and Dispute Resolution for Freelancers

Complex Computing Problem – Semester Project Proposal | Name: Zain ul arfin | Roll No: 054 | Supervisor: [Name]

1. Introduction

Freelancing depends on trust between strangers. Freelancers risk unpaid work and clients risk paying for work that is not delivered. This project proposes FreelanceSafe Lite, a Java web application in which payments are held in milestone-based escrow (simulated), accounts and contracts receive a simple rule-based risk score, and disputes are resolved through evidence submitted by both sides and a transparent recommendation that an administrator reviews.

2. Problem Statement

Escrow, risk control, disputes and reputation depend on each other. Stricter risk checks protect users but may block honest newcomers; faster decisions reduce waiting but may be less fair; collecting evidence helps fairness but must respect privacy. The problem has no single optimal policy, so the project analyzes trade-offs between these goals using clear rules and test scenarios.

3. Objectives

- Implement milestone-based escrow with controlled states and simulated payments.
- Provide role-based access for clients, freelancers and administrators.
- Calculate a rule-based risk score from account and payment signals.
- Support disputes with evidence from both parties and a weighted recommendation reviewed by an administrator.
- Maintain a reputation score updated after each completed contract.
- Test that invalid payment actions are blocked and compare results with manual handling without scoring.

4. Proposed System

Component	Responsibility
Escrow Engine	Manages milestone states (Created, Funded, Submitted, Released, Disputed, Resolved) using the State pattern and simulated payments.
Risk Scoring	Weights signals such as new account, high amount, repeated refunds and mismatched details; the scoring rules use the Strategy pattern so they can be compared.
Dispute Module	Collects evidence (messages, files, milestone history) from both sides and recommends an outcome; the administrator makes the final decision.
Reputation	Updates trust scores from completed contracts, disputes and ratings.
Dashboard and Audit	Role-based pages, alerts and an audit log of every state change.

5. Justification as a Complex Computing Problem

Characteristic	Project-specific evidence
Conflicting Requirements	Security vs ease of use for new users; fast decisions vs fairness and human review; evidence collection vs privacy.

Characteristic	Project-specific evidence
Stakeholder Involvement	Clients, freelancers and administrators have different permissions, goals and risks; for example only a client can release funds.
Interdependence	Escrow state affects risk scoring, risk flags affect releases, dispute outcomes affect reputation, and reputation affects future risk scores.
Consequences	Wrong payment or dispute decisions cause financial loss, unfair outcomes and damaged reputations. High-impact outcomes require administrator review and appeal.
Depth of Analysis (supporting)	Scoring rules and thresholds, and admin-only vs weighted-evidence dispute handling, are compared using false-block rate, missed-risk rate and dispute agreement.

6. Methodology

- Implementation: Java 17 with Spring Boot (layered Controller, Service and Repository design), Spring Security for authentication and roles, JPA/Hibernate with MySQL, and a Thymeleaf web dashboard.
- Design patterns: State for escrow transitions, Strategy for risk scoring, Observer for reputation and audit updates.
- Payments are simulated; no real card or bank data is used.
- Testing: JUnit and Mockito unit tests, plus hand-made scenarios.

7. Evaluation Plan

Scenarios: honest transaction, client non-payment, fake refund claim and invalid state change. Metrics: false-block rate, missed-risk rate, dispute outcome agreement with an expected answer, and resolution time. Results are compared with manual dispute handling without scoring.

8. Timeline

Weeks	Activity
1–3	Requirements, literature review, data collection
4–6	Java architecture, escrow engine, test data
7–9	Risk score, dispute and reputation logic
10–12	Dashboard, integration and testing
13–14	Evaluation, final report and demonstration

9. Risks and Ethical Considerations

- Only simulated and test data are used; no real payment or personal data.
- Rule-based scores may unfairly affect new freelancers, so false-block rates are measured and thresholds adjusted.
- Recommendations are advisory; an administrator decides, and either party can appeal.

10. Expected Deliverables

- Working prototype with dashboard and decision explanations.
- Comparative analysis of scoring rules and dispute handling, with trade-off discussion.
- Final report, presentation and demonstration.